Venkata Lakshmi Parimala Pasupuleti

AI / LLM Engineer · Multi-Agent Orchestration · LangGraph · MCP · RAG · Vector Search · LLM Evaluation · Guardrails · Python · AWS

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SUMMARY

AI/LLM engineer with 9+ years building production systems, including 2+ years building and shipping LLM-based applications. Built **Leasel** - a production multi-step, tool-using agent platform: **8-agent LangGraph orchestration** with planning, memory, retrieval, and conditional routing; **MCP server integration** for shared tool access; **hybrid search RAG** (OCR ingestion, ChromaDB vector search, TF-IDF + semantic reranking, citation grounding for accuracy); **HITL guardrails** for safety; **CI eval harness** (LLM-as-judge, 93.6% precision, 1,240 test cases). Built **AgentProbe** - an open-source LLMOps and evaluation platform: LLM-as-judge scoring, CI regression gating, cost-aware model routing across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama, and full per-run observability and tracing. Currently at Intuit integrating Anthropic Claude into production financial services: autonomous agent workflows, prompt/context engineering, structured outputs, RAG-based retrieval, guardrails, and LLM evaluation harnesses with CI gates. Deep familiarity with LLM APIs (Anthropic, OpenAI, Gemini, open-weight models), **fine-tuning** (HuggingFace + PyTorch, custom classification model), async distributed systems, Docker, Kubernetes, AWS. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Multi-Step Tool-Using Agents · Planning · Memory · Retrieval · Orchestration

LangGraph · LangChain · Agent SDKs · CrewAI-compatible Patterns

MCP · Model Context Protocol · Function Calling · Tool Integrations

RAG Pipelines · Vector Search · ChromaDB · pgvector · Embeddings

Context Management · Prompt Engineering · Structured Outputs

LLM Evaluation · LLM-as-Judge · Safety · Reliability · Probabilistic Systems

LLMOps · Observability · Tracing · CI Regression · Model Routing

Guardrails · HITL · Failure Handling · Input Validation

LLM APIs · Anthropic · OpenAI · Gemini · Open-Weight Models · AWS Bedrock

Fine-Tuning · HuggingFace · PyTorch · RLHF-adjacent Techniques

Python · FastAPI · TypeScript · Async Systems · Distributed Services

AWS · Docker · Kubernetes · PostgreSQL · CI/CD

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA (Remote)

- **Production LLM agent development:** Architect and build LangChain-based autonomous agent workflows integrating Anthropic Claude - multi-step orchestration with tool/function calling, prompt/context engineering, structured-output schemas, RAG-based retrieval grounding, and MCP-style tool integrations; guardrails with input validation, sensitive-data handling, and fallbacks; HITL for safety-critical decisions.
- **LLM evaluation and LLMOps:** Build LLM-as-judge eval frameworks with golden datasets and CI regression gates; monitor production AI via structured logging and CloudWatch quality signals; perform error analysis on failures and convert to eval coverage. Build prompt management patterns and playbooks.
- Build ML data pipelines (scikit-learn, XGBoost); React/TypeScript front ends; deploy via Docker + GitHub Actions CI/CD to AWS.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Apartment & Affordable Housing SaaS (regulated multi-tenant, 100K+ monthly active users)

- Owned mission-critical async distributed systems end-to-end - HUD compliance ETL (180K records, zero violations across 6 years), RabbitMQ event-driven processing with idempotency guarantees, PHI-compliant AWS S3 migration (100% coverage, zero incidents).
- Led Docker microservices redesign (50% scalability improvement); optimized 200+ SQL queries (30% performance gain); Python/Bash automation eliminating 80% of manual workload; mentored 5+ engineers; 300+ code reviews across 120+ Agile sprints.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration workflows (1.6M records); led AWS S3 migration from legacy storage.

PROJECTS

Leasel - Multi-Agent Platform: LangGraph · MCP · RAG · Vector Search · Guardrails

2026 | Python, LangGraph (LangChain), FastAPI, ChromaDB, MCP, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production multi-step, tool-using agent platform: **8-agent LangGraph orchestration** with planning, memory, tool/function calling, and conditional routing; **MCP server integration** for shared tool access across agents; **RAG pipeline** (OCR ingestion, ChromaDB vector search, TF-IDF + semantic reranking, citation grounding for accuracy and cost efficiency); **HITL guardrails** for safety-critical decisions; **CI eval harness** with LLM-as-judge and golden datasets: 93.6% precision, 87.9% F1 across 1,240 test cases.

AgentProbe - LLMOps & Evaluation Platform: Model Routing · CI Regression · Observability

2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source **LLMOps and evaluation platform**: offline golden-dataset evals, LLM-as-judge scoring, CI regression harness gating releases on eval thresholds, failure handling with structured error classification; **cost-aware model routing** across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama; full per-run tracing and structured logging. Covers LangSmith/Langfuse use cases for self-hosted LLMOps.

AI on FHIR — Conversational AI & API Integration

2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates natural-language questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut clinical lookup time ~60%. Third-party API integration, OAuth-secured, HIPAA-style RBAC, Docker-containerized.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017